

REGION 1

Set up all six orders of integration for

$$\iiint_{R_1} f(x, y, z) dV$$

where R_1 is the region bounded by the planes

$$y = 0, z = 0, x = 2, y = x, \text{ and } z = x + y$$

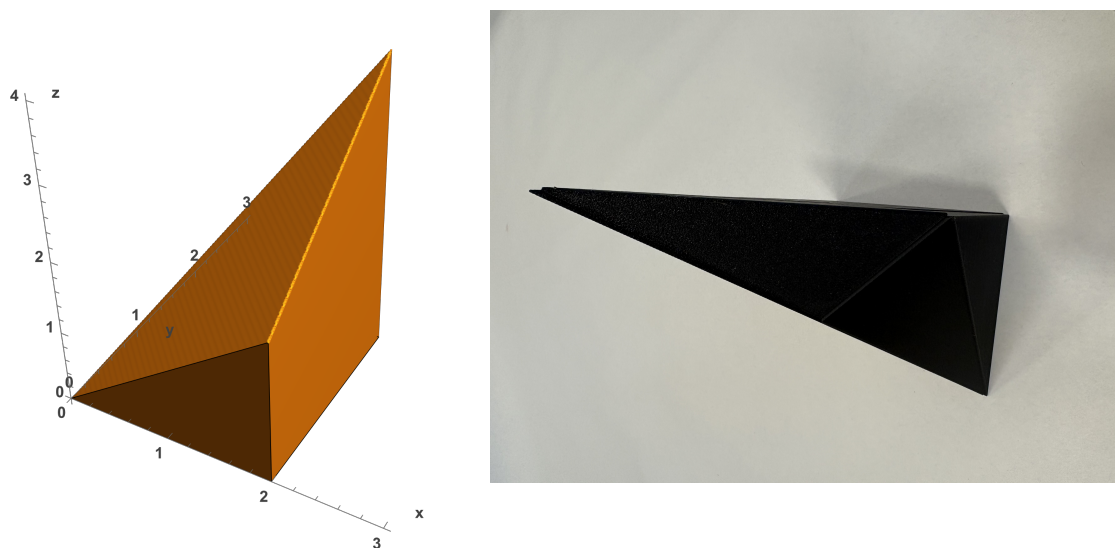


FIGURE 1. Computer-modeled and 3D-printed views of R_1 .

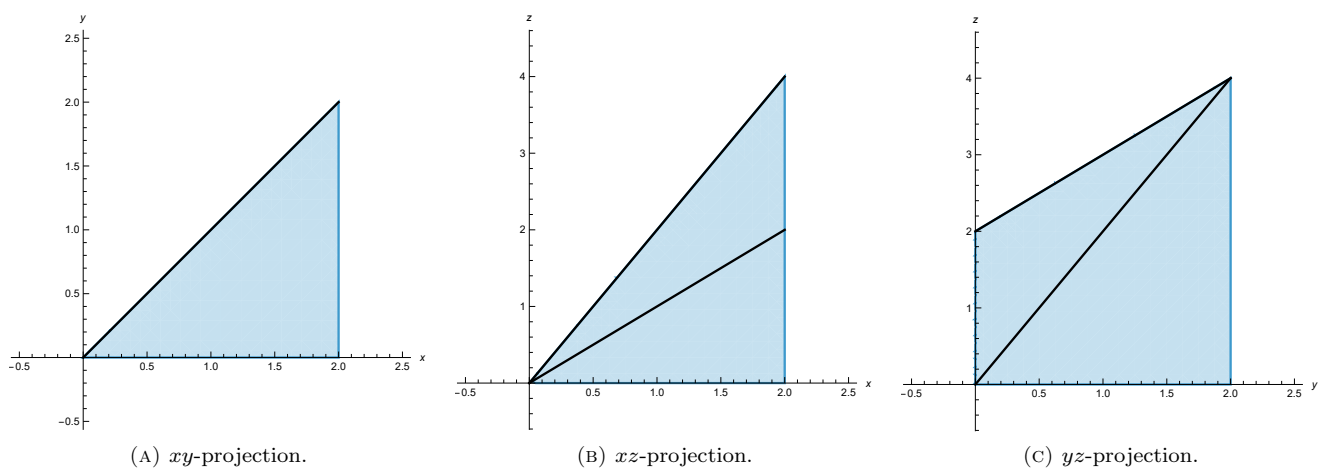


FIGURE 2. Projections of R_1 into the three coordinate planes.

The following triple integrals represent integrating over the region using the xy -projection:

$$\int_0^2 \int_0^x \int_0^{x+y} f(x, y, z) \, dz \, dy \, dx$$

$$\int_0^2 \int_y^2 \int_0^{x+y} f(x, y, z) \, dz \, dx \, dy$$

The following triple integrals represent integrating over the region using the xz -projection:

$$\int_0^2 \int_0^x \int_0^x f(x, y, z) \, dy \, dz \, dx + \int_0^2 \int_x^{2x} \int_{z-x}^x f(x, y, z) \, dy \, dz \, dx$$

$$\int_0^2 \int_z^2 \int_0^x f(x, y, z) \, dy \, dx \, dz + \int_0^2 \int_{\frac{z}{2}}^z \int_{z-x}^x f(x, y, z) \, dy \, dx \, dz + \int_2^4 \int_{\frac{z}{2}}^2 \int_{z-x}^x f(x, y, z) \, dy \, dx \, dz$$

The following triple integrals represent integrating over the region using the yz -projection:

$$\int_0^4 \int_{\frac{z}{2}}^2 \int_y^2 f(x, y, z) \, dx \, dy \, dz + \int_0^2 \int_0^{\frac{z}{2}} \int_{z-y}^2 f(x, y, z) \, dx \, dy \, dz + \int_2^4 \int_{z-2}^{\frac{z}{2}} \int_{z-y}^2 1 \, dx \, dy \, dz$$

$$\int_0^2 \int_0^{2y} \int_y^2 f(x, y, z) \, dx \, dz \, dy + \int_0^2 \int_{2y}^{y+2} \int_{z-y}^2 f(x, y, z) \, dx \, dz \, dy$$

All of these triple integrals evaluate to 4 when $f(x, y, z) = 1$.